

Augmented Reality for Warehouse: Aid System for Foreign Workers

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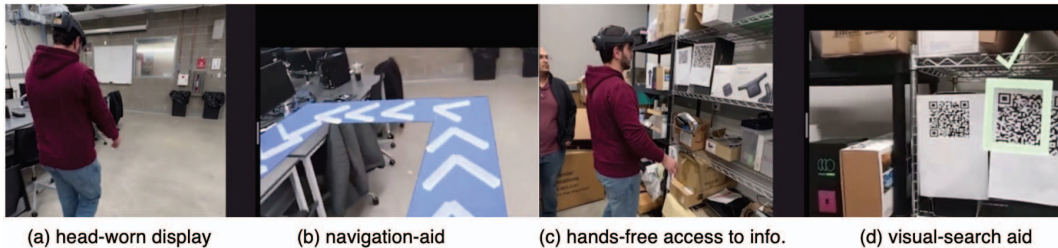


Figure 1: The proposed augmented reality warehouse order-picking aid uses a head-worn display (a) and directly overlays visual symbols atop the real-world task environment to (b) guide the user to the right pallet. It also allows hands-free access (c) to contextual information to help the user (d) identify the target package.

ABSTRACT

Order picking in inventory warehouses is one of the challenging tasks, especially for foreign workers due to language barriers found in current order picking instructions. To address this challenge, we took the user-centered design approach and developed an augmented reality (AR) aid system using a head-worn display (HWD). Unlike most AR warehouse aid systems that present icons or texts on fixed locations of the displays, we directly overlaid visual symbols atop the real-world to better guide foreign workers' tasks. Our pilot user study demonstrated the potential of the proposed system in improving user performance, perceived safety, and usability as compared to one of the current practices for order picking, verbal instructions. This work highlights the importance of the user-centered design of AR solutions that bridge the gap between end-users and available technologies in warehouses. The proposed AR job aid system could be developed further to support more complex warehouse tasks.

Keywords: Augmented reality, Human-computer interaction, User interface application, Warehouse.

Index Terms: H.5 [Information Interfaces and Presentation]: H.5.1: Multimedia Information Systems — Artificial, Augmented, and Virtual Realities

1 INTRODUCTION

Warehouse environments store a variety of products with different packaging. A series of tasks for locating, verifying, and moving packages for shipments is commonly known as retrieving packages or order-picking, is one of the most important warehouse processes as Tompkins estimated the cost of it as 55% of the warehouse operating cost [1]. Continuous changes occurring in the

order-picking process make human operators' contribution to the job necessary in many cases since manual picking offers high flexibility [2]. However, operators may often lose focus and get confused with job instructions which end up getting lost while looking for the right pallet that needs to be located. It is also common for operators to pick up the wrong packages and make the wrong shipments to the customers. This problem gets further complicated in maintaining inventories of several components being shipped to different customers. Especially for foreign workers who have different language backgrounds, existing order-picking technologies such as voice-directed and paper-based systems add even more challenges.

Augmented reality (AR) has the potential to address these problems by overlaying contextual information on the real-world task environment and intuitively guiding the operator's attention to the task at hand.

2 OBJECTIVE

Given the potential of AR HWDs and challenges in warehouse order picking tasks, this work aimed to develop AR aid to enhance foreign workers' experience and performance.

3 METHOD

To bridge the gap between the end-users and the available technologies in warehouses, we took the user-centered design approach to the problem proposed by Hartson & Pyla [3], which consists of understanding, design, prototyping, and evaluation phases.

We interviewed 5 warehouse workers to better understand the current practices for order-picking in warehouses. Interviews include questions related to the current inventory order-picking process and the language skills that may affect worker performance. Our interviews resulted in key findings: It is often challenging for workers to correctly understand verbal instructions due to the high noise in the warehouse environment and language barriers. This challenge often resulted in picking errors and had negative consequences on the operation of the warehouse.

We design and prototype AR user interfaces (UI) based on user personas and storyboard we developed. We also critiqued our initial UI design alternatives relying on Ganapathy eight design guidelines [4] for mobile AR to make the prototype more involved with personas with more noticeable and meaningful navigation signs regarding directions (Figure 1). A prototype has been developed using unreal engine 4.27 and Microsoft Mixed Reality

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Toolkit plugins based on the design ideas to produce an application to be used with Microsoft HoloLens 2.

We conducted a pilot user study involving 6 participants from different languages background who use English as their second language. Participants were asked to (1) identify a package to be picked up from a work order at a base-station, (2) move to the shelf that stores the target package, (3) locate the target package, (4) bring it to the base station. Each participant completed the task twice with and without the AR aid system. In the without AR aid condition, participants relied on verbal instructions given by a researcher. To quantify the effects of the AR aid system on user performance and experience, after each trial, we collected data from participants relying on three main constructs: (1) system usability, (2) safety, and (3) performance. For the system usability, we used the usability scale (SUS) [5]. For safety we developed a questionnaire to measure the effect of the proposed system on users' sensation, perception, and awareness of any potential risks. Finally, for performance we choose to measure task completion time and error rate.

4 RESULTS

The pilot study found that the proposed AR aid system didn't have effects on task completion time, but it did improve the error rate by decreasing it to zero with the AR aid system comparing to 16% without the aid system. Five out of six participants agreed that using the AR headwear didn't affect their sensation, perception, reaction, and awareness of any potential risks while one participant indicated that the headset affected the vision by 20%, danger sensation by 40% and reaction by 60%, though the participant was able to avoid all potential risks in the environment.

5 CONCLUSION

Although the performance was improved only in error rate aspect the primary focus of the application of designing an AR solution for users from different languages background to enhance their performance and experience was achieved using our proposal. We may get more performance enhancement when users get more familiar with the system as the comparison was against the common traditional verbal instructions that all participants are familiar with.

6 IMPLICATIONS AND FUTURE WORK

The results of this study indicate the potential of AR job aids in improving worker performance and experience. The proposed system could be further improved for more complicated warehouses where human operators' awareness of the dynamic environment is critical for worker safety. The research findings and proposed methods could be extended to other safety-critical domains (e.g., transportation, construction, and military) where human operators' awareness of the dynamic environment is important [6].

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